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Kinetic modeling for transport of elastic waves in anisotropic heterogeneous media

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Abstract

This paper deals with the derivation of transport equations for high-frequency elastic bulk waves propagating in anisotropic heterogeneous media. The former describe the evolution of the energy density associated to those waves in the phase space position $\times$ wave vector. The model accounts for all possible polarization of waves in an anisotropic elastic medium and their interactions, as well as for the degeneracy directions of propagation when two phase speeds possibly coincide on acoustic axes. Thus it embodies isotropic elasticity which was considered in several previous publications. The derivation is based on a multi-scale asymptotic analysis using spatio-temporal Wigner transforms and their interpretation in terms of semiclassical operators.

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1. Introduction

Electromagnetic, acoustic or elastic waves in a homogeneous medium have a fairly well identified behavior, even in the presence of a smooth boundary as a free surface for example. This is no longer the case in a heterogeneous medium or a piecewise homogeneous medium, not to mention a random medium. Wave propagation in heterogeneous media is the subject of intense, multidisciplinary researches in order to exhibit, on one hand, some new phenomena such as anisotropic diffusion (strong localization) or coherent back-scattering effects (weak localization), and to take advantage, on the other hand, of these effects in applications such as time reversal techniques [1, 2]. This issue is less studied in continuum mechanics, beside some early works in seismology and surface geophysics [3, 4]. However it is essential in the understanding of high-frequency wave propagation phenomena in elastic media at the mesoscopic and macroscopic scales, for applications in structural-acoustics [5, 6], room acoustics [7], or microstructural characterization of polycrystalline materials by ultrasonic waves [8], among others.

Introducing the wavelength λ and a characteristic length of the heterogeneities ℓ , such as a correlation length, the

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microscopic, mesoscopic and macroscopic regimes of wave propagation are classically defined by $\delta = \lambda/\ell > 1$, $\delta \approx 1$, and $\delta < 1$ respectively [1]. In the microscopic regime the relevant evolution model is the wave equation, requiring *a priori* a precise knowledge of the shape and location of the heterogeneities. The limit case $\delta \gg 1$ where the mechanical parameters of the medium vary rapidly can be treated by homogenization techniques or an effective medium theory [9, 10]. Both approaches extend to the case of randomly distributed inclusions of which number increases indefinitely at a constant volume fraction in a homogeneous background medium. In the mesoscopic regime, the up-scaled evolution model is the transport equation, or radiative transfer (linear Boltzmann) equation in a high-frequency random medium where the correlation length is comparable to the wavelength [11, 12]. It corresponds to a situation of strong interaction between waves and random heterogeneities which can not be addressed by usual homogenization and multi-scale techniques. The characteristic homogenized parameters of this medium are the scattering mean free paths and transport velocities, which are derived from the microscopic regime. At last in the macroscopic regime the relevant evolution model is the diffusion equation [5]. It is characterized by a diffusion operator which is derived from the mesoscopic parameters. In this regime all interferences of waves with the heterogeneities are smoothed out and angularly averaged wave speckles are considered.

The objective of the research initiated in this paper is to assess the influence of material anisotropy on the mesoscopic regime of elastic waves in randomly heterogeneous media. Anisotropy should be considered at two levels. The first one is related to the constitutive equation of random materials, which may be handled by a random matrix theory for the elasticity tensor [13, 14]. The second level is related to the correlation structure of these random materials, which is referred to as anisotropy in the dedicated literature. Since the propagation of waves in such complex media cannot be described by deterministic models as already outlined above, a so-called kinetic framework shall be adopted. It is based on a transport, or radiative transfer model coupling all different polarization modes of the elastic waves. Transport and radiative transfer equations can be derived from the elastic wave equation by a classical micro-local analysis based on rigorous mathematical arguments [15, 16]. The main purpose of the paper is to extend those results to anisotropic elasticity.

The rest of the paper is organized as follows. Sect. 2 recalls the basic framework of elastic wave propagation in anisotropic media and introduces the high-frequency (small wavelength) scaling. The high-frequency limit is considered in Sect. 3 where the transport equations for the scalar and matrix energy densities associated to each polarization of the waves are derived. The constitutive parameters of the medium are assumed to vary at the large scale $O(1)$ with respect to the small wavelength. The elements of semiclassical calculus used in the analysis are also recalled in this section. In Sect. 4 the high-frequency limit is outlined for constitutive parameters varying this time at the same small scale as the wavelength, a regime of expectedly maximum interactions between the waves and the medium. Sect. 5 summarizes these results and proposes some directions of future researches.

2. Elastic bulk wave propagation in a high-frequency setting

2.1. Elastic wave equation

We first consider the derivation of the wave equation in an open domain $\mathcal{O} \subseteq \mathbb{R}^3$ occupied by a heterogeneous linear viscoelastic material, where $\mathbb{R}^3$ is the three-dimensional Euclidean space. Its density is denoted by $\rho(\mathbf{x})$ and its fourth-order, time-dependent relaxation tensor is denoted by $\mathbf{C}(\mathbf{x}, t)$, with $\mathbf{x} \in \mathcal{O}$ and $t \in \mathbb{R}$. Let $\mathbf{u}_\varepsilon(\mathbf{x}, t) \in \mathbb{R}^3$ (the subscript ε will be made clear just below) be the displacement field of that medium about a quasi-static equilibrium considered as the reference configuration, and $\boldsymbol{\sigma}_\varepsilon(\mathbf{x}, t)$ be its second-order Cauchy stress tensor. The balance of momentum in a fixed reference frame ignoring the action of body forces is:

$$\rho \partial_t^2 \mathbf{u}_\varepsilon = \text{Div } \boldsymbol{\sigma}_\varepsilon, \quad \mathbf{x}, t \in \mathcal{O} \times \mathbb{R}. \quad (1)$$

Here the divergence of a second-order tensor $\mathbf{A}$ is defined by $(\text{Div } \mathbf{A}, \mathbf{b}) = \nabla_{\mathbf{x}} \cdot (\mathbf{A}^T \mathbf{b})$ for any constant vector $\mathbf{b}$, and $\nabla_{\mathbf{x}}$ is the gradient vector with respect to the point $\mathbf{x} \in \mathbb{R}^d$. At last $(\cdot)^T$ stands for the matrix transpose. Eq. (1) is supplemented by the initial conditions:

$$\mathbf{u}_\varepsilon(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x}; \varepsilon), \quad \partial_t \mathbf{u}_\varepsilon(\mathbf{x}, 0) = \mathbf{v}_0(\mathbf{x}; \varepsilon), \quad (2)$$

which are parameterised by a small scaling parameter, say ε . It quantifies the rate of change of the initial state $\mathbf{u}_0$ and $\mathbf{v}_0$ with respect to the dimensions of $\mathcal{O}$ or the propagation/observation distances. Since high-frequency waves

will be generated by an initial vibrational energy oscillating at a scale $\varepsilon \ll 1$, the initial conditions $\nabla_{\mathbf{x}} \otimes \mathbf{u}_0$ and $\mathbf{v}_0$ shall be considered as strongly ε -oscillating functions [11, 15]. The plane waves $\mathbf{u}_0(\mathbf{x}; \varepsilon) = \varepsilon \mathbf{A}(\mathbf{x}) e^{i\mathbf{k} \cdot \mathbf{x} / \varepsilon}$ and $\mathbf{v}_0(\mathbf{x}) = \mathbf{B}(\mathbf{x}) e^{i\mathbf{k} \cdot \mathbf{x} / \varepsilon}$ for a given wave vector $\mathbf{k} \in \mathbb{R}^3$ and $i = \sqrt{-1}$, typically fulfill this condition. Then the displacement field $\mathbf{u}_\varepsilon$ is ε -oscillatory as well, ignoring boundary effects at first glance.

The stress field $\sigma_\varepsilon(\mathbf{x}, t)$ on $\bar{\mathcal{O}} \times \mathbb{R}$ is given as a function of the linearised strain tensor $\varepsilon_\varepsilon(\mathbf{x}, t)$ by the material constitutive equation:

$$\sigma_\varepsilon(\mathbf{x}, t) = \int_{-\infty}^t \mathbf{C}(\mathbf{x}, t - \tau) \partial_t \varepsilon_\varepsilon(\mathbf{x}, \tau) d\tau, \quad \varepsilon_\varepsilon(\mathbf{x}, t) = \nabla_{\mathbf{x}} \otimes_s \mathbf{u}_\varepsilon(\mathbf{x}, t). \quad (3)$$

Both σ_ε and ε_ε are parameterised by the scale ε of the applied loads since $\mathbf{u}_\varepsilon$ also is. Here $\otimes_s$ is the symmetrised tensor product of two vectors $\mathbf{a} \otimes_s \mathbf{b} = \text{sym}(\mathbf{a} \otimes \mathbf{b})$. If the motion starts at time $t = 0$, and if $\varepsilon_\varepsilon = \sigma_\varepsilon = \mathbf{0}$ for $t < 0$, it can be shown (by integration by parts) that the constitutive equation becomes:

$$\sigma_\varepsilon(\mathbf{x}, t) = \mathbf{C}(\mathbf{x}, 0) \varepsilon_\varepsilon(\mathbf{x}, t) + \int_0^t \partial_t \mathbf{C}(\mathbf{x}, \tau) \varepsilon_\varepsilon(\mathbf{x}, t - \tau) d\tau \quad (4a)$$

$$= \partial_t \left(\int_0^t \mathbf{C}(\mathbf{x}, \tau) \varepsilon_\varepsilon(\mathbf{x}, t - \tau) d\tau \right), \quad \forall t \geq 0, \quad \forall \mathbf{x} \in \bar{\mathcal{O}}. \quad (4b)$$

It may be shown in addition [17] that the convolution integral in the first equality above becomes ineffective in the high-frequency limit $\varepsilon \rightarrow 0$ that will be considered in the subsequent sections. This holds true as long as the relaxation tensor $\mathbf{C}$ remains independent of ε , which is assumed here and in the subsequent Sect. 3. It should be noted however that this theoretical result is in basic agreement with the current observations: viscoelastic materials behave as purely elastic materials when they are forced by high-frequency harmonic motions, see *e.g.* [18]. Therefore a linear elastic constitutive equation $\sigma_\varepsilon(\mathbf{x}, t) = \mathbf{C}(\mathbf{x}, 0) \varepsilon_\varepsilon(\mathbf{x}, t)$ is adopted in the remaining of the paper. The elastic wave equation is derived plugging this relation into Eq. (1) and thus reads:

$$\rho \partial_t^2 \mathbf{u}_\varepsilon - \nabla_{\mathbf{x}} \cdot (\mathbf{C} : \nabla_{\mathbf{x}} \otimes \mathbf{u}_\varepsilon) = \mathbf{0}, \quad \mathbf{x}, t \in \mathcal{O} \times \mathbb{R}, \quad (5)$$

where $\mathbf{C}$ may be seen as a symmetric elasticity tensor independent of time.

2.2. The Christoffel tensor

The acoustic, or Christoffel tensor $\Gamma(\mathbf{x}, \mathbf{k})$ of the propagation medium is defined by:

$$\Gamma(\mathbf{x}, \mathbf{k}) \mathbf{U} = \rho(\mathbf{x})^{-1} (\mathbf{C}(\mathbf{x}) : \mathbf{U} \otimes \mathbf{k}) \mathbf{k}, \quad \mathbf{U} \in \mathbb{R}^3, \mathbf{k} \in \mathbb{R}^3. \quad (6)$$

It is symmetric, real and positive definite in $\mathcal{O} \times \mathbb{R}^3 \setminus \{\mathbf{k} = \mathbf{0}\}$. So for a given direction $\hat{\mathbf{k}} := \mathbf{k}/|\mathbf{k}|$ on the unit sphere $\mathbb{S}^2$ of $\mathbb{R}^3$ it has three real positive (possibly multiple) eigenvalues $c_\alpha^2(\mathbf{x}, \mathbf{k})$ for $\alpha = 1, 2$ or 3 , and the associated eigenvectors $\mathbf{p}_\alpha(\mathbf{x}, \mathbf{k})$ can be chosen real and orthogonal. They can also be normalized such that:

$$\Gamma(\mathbf{x}, \mathbf{k}) = \sum_{\alpha=1}^3 c_\alpha^2 \mathbf{p}_\alpha \otimes \mathbf{p}_\alpha, \quad \mathbf{I} = \sum_{\alpha=1}^3 \mathbf{p}_\alpha \otimes \mathbf{p}_\alpha, \quad (7)$$

where $\mathbf{I}$ is the identity matrix of $\mathbb{R}^3$. The eigenvalues and eigenvectors of the Christoffel tensor correspond to the phase velocities and polarizations of plane waves propagating along the direction $\hat{\mathbf{k}}$ in a medium characterized by its elasticity tensor $\mathbf{C}$. The polarization with the closest direction to $\hat{\mathbf{k}}$ is called the quasi-longitudinal wave, the other two components being the quasi-transversal waves. The above spectral expansion of the Christoffel tensor is valid for all directions but the so-called acoustic axes, along which two eigenvalues may coincide. For such a direction $\hat{\mathbf{k}} = \hat{\mathbf{k}}_a$ of degeneracy where, say $c_1(\mathbf{x}, \hat{\mathbf{k}}_a) = c_2(\mathbf{x}, \hat{\mathbf{k}}_a)$, the expansion reduces to:

$$\Gamma(\mathbf{x}, \hat{\mathbf{k}}_a) = c_1^2(\mathbf{x}, \hat{\mathbf{k}}_a) \mathbf{I} + (c_3^2(\mathbf{x}, \hat{\mathbf{k}}_a) - c_1^2(\mathbf{x}, \hat{\mathbf{k}}_a)) \mathbf{p}_3(\mathbf{x}, \hat{\mathbf{k}}_a) \otimes \mathbf{p}_3(\mathbf{x}, \hat{\mathbf{k}}_a). \quad (8)$$

It follows from Eq. (8) that any vector which is orthogonal to $\mathbf{p}_3(\mathbf{x}, \hat{\mathbf{k}}_a)$ is an eigenvector of the acoustic tensor $\Gamma(\mathbf{x}, \hat{\mathbf{k}}_a)$, *i.e.* it is an allowed polarization for a wave propagating along $\hat{\mathbf{k}}_a$ with a wave celerity $c_1(\mathbf{x}, \hat{\mathbf{k}}_a)$. A common example

is elastic isotropy, when $\mathbf{C} = \lambda \mathbf{I} \otimes \mathbf{I} + 2\mu \mathbf{I} \boxtimes \mathbf{I}$ where λ and μ are the Lamé parameters and $(\mathbf{A} \boxtimes \mathbf{A})\mathbf{B} := \mathbf{A} \text{sym}(\mathbf{B})\mathbf{A}$. Then the Christoffel tensor reads:

$$\Gamma(\mathbf{x}, \hat{\mathbf{k}}) = c_S^2(\mathbf{x})\mathbf{I} + (c_P^2(\mathbf{x}) - c_S^2(\mathbf{x}))\hat{\mathbf{k}} \otimes \hat{\mathbf{k}}, \quad \forall \hat{\mathbf{k}} \in \mathbb{S}^2, \quad (9)$$

where $c_P = \sqrt{(\lambda + 2\mu)/\rho}$ and $c_S = \sqrt{\mu/\rho}$ are the velocities for compressive and shear waves, respectively. The various properties of the Christoffel tensor for anisotropic media are discussed in *e.g.* [19] and references therein.

2.3. High-frequency setting

The high-frequency limit $\varepsilon \rightarrow 0$ will be characterized by considering quadratic quantities such as the vibrational strain energy or the kinetic energy in this very limit, because of the following reasons. At first, let us consider a real function $x \mapsto u_\varepsilon(x)$ oscillating with an amplitude $a(x)$ about its mean $\underline{u}(x)$ both functions being slowly varying, for example:

$$u_\varepsilon(x) = \underline{u}(x) + a(x) \sin \frac{x}{\varepsilon}, \quad 0 < \varepsilon \ll 1. \quad (10)$$

This function has no strong limit when $\varepsilon \rightarrow 0$. However if its vague limit is of interest, namely:

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(x) (u_\varepsilon(x))^2 dx$$

for any continuous function φ with compact support on $\mathbb{R}$, it is found that:

$$\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(x) (u_\varepsilon(x))^2 dx = \int_{\mathbb{R}} \varphi(x) \left((\underline{u}(x))^2 + \frac{1}{2} (a(x))^2 \right) dx. \quad (11)$$

Thus the "energy" associated with $u_\varepsilon(x)$ admits, locally at any point, a limit given by $\underline{u}(x)^2 + \frac{1}{2}a(x)^2$; see Fig. 1 below. It allows to "see" the deviation of oscillations with amplitude $a(x)$ about the mean at any point x selected by the observation function φ , whereas the weak limit of $u_\varepsilon(x)$ does not. At last it no longer has the oscillatory feature of the initial function, a simplification which is of high interest for experiments and simulations.

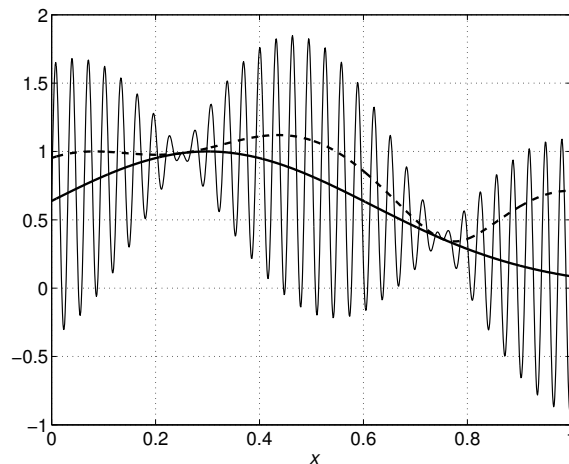


Fig. 1. The energy limit of a strongly oscillating sequence. Thin line: the oscillating function $u_\varepsilon(x)$, thick line: the mean function $\underline{u}(x)$, and thick dashed line: the square-root limit $(\underline{u}(x)^2 + \frac{1}{2}a(x)^2)^{\frac{1}{2}}$. After [6].

The notion of semiclassical measure, or Wigner measure [11, 15, 16] can be seen as a mathematical generalisation of these concept and ideas. Let φ be a so-called 3×3 matrix observable of which coefficients are smooth, compactly supported functions defined on $T^*\mathbb{R}^3 \equiv \mathbb{R}_x^3 \times \mathbb{R}_k^3$. For a square integrable vector field $\mathbf{u} \in [L^2(\mathbb{R}^3)]^3$, the functional

space of $\mathbb{R}^3$ -valued, square integrable functions equipped with the scalar product $(\mathbf{u}, \mathbf{v})_{L^2} = \int_{\mathbb{R}^3} \mathbf{u}(\mathbf{x}) \cdot \mathbf{v}(\mathbf{x}) \, d\mathbf{x}$, consider the (semiclassical) operator:

$$\varphi(\mathbf{x}, \varepsilon \mathbf{D}) \mathbf{u}(\mathbf{x}) = \varphi(\mathbf{x}, \varepsilon \mathbf{D}) \left(\int_{\mathbb{R}^3} e^{i\mathbf{k} \cdot \mathbf{x}} \widehat{\mathbf{u}}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^3} \right) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i\mathbf{k} \cdot \mathbf{x}} \varphi(\mathbf{x}, \varepsilon \mathbf{k}) \widehat{\mathbf{u}}(\mathbf{k}) \, d\mathbf{k}, \quad (12)$$

where $\widehat{\mathbf{u}}(\mathbf{k}) := \int_{\mathbb{R}^3} e^{-i\mathbf{k} \cdot \mathbf{x}} \mathbf{u}(\mathbf{x}) \, d\mathbf{x}$ stands for the Fourier transform of $\mathbf{u}(\mathbf{x})$. Then for a sequence $(\mathbf{u}_\varepsilon)$ which is uniformly bounded in $[L^2(\mathbb{R}^3)]^3$, there exists a positive, Hermitian measure $\mathbf{W}[\mathbf{u}_\varepsilon]$ such that, up to extracting a subsequence if need be:

$$\lim_{\varepsilon \rightarrow 0} (\varphi(\mathbf{x}, \varepsilon \mathbf{D}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} = \text{Tr} \int_{T^*\mathbb{R}^3} \varphi(\mathbf{x}, \mathbf{k}) \mathbf{W}[\mathbf{u}_\varepsilon](d\mathbf{x}, d\mathbf{k}), \quad \forall \varphi. \quad (13)$$

$\mathbf{W}[\mathbf{u}_\varepsilon]$ is the so-called Wigner measure of the sequence $(\mathbf{u}_\varepsilon)$ because it can also be interpreted as the weak limit of its Wigner transform $\mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)$. Indeed, if the latter is defined for $\mathbb{R}^3$ -valued temperate distributions $\mathbf{u}, \mathbf{v}$ by:

$$\mathbf{W}_\varepsilon(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{k}) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i\mathbf{k} \cdot \mathbf{y}} \mathbf{u}(\mathbf{x} - \varepsilon \mathbf{y}) \otimes \overline{\mathbf{v}(\mathbf{x})} \, d\mathbf{y}, \quad (14)$$

then one has:

$$(\varphi(\mathbf{x}, \varepsilon \mathbf{D}) \mathbf{u}, \mathbf{v})_{L^2} = \text{Tr} \int_{T^*\mathbb{R}^3} \varphi(\mathbf{x}, \mathbf{k}) \mathbf{W}_\varepsilon(\mathbf{u}, \mathbf{v})(d\mathbf{x}, d\mathbf{k}).$$

Thus $\mathbf{W}[\mathbf{u}_\varepsilon]$ describes the limit energy of the sequence $(\mathbf{u}_\varepsilon)$ in the phase space $T^*\mathbb{R}^3$. As in Eq. (11), the matrix function $\varphi(\mathbf{x}, \mathbf{k})$ is used to select any quadratic observable or quantity of interest associated to this energy: the kinetic energy, or the free energy, or the power flow, etc. For example, the high-frequency strain energy $\mathcal{V}_\varepsilon(t) := \frac{1}{2} \int_{\mathcal{O}} \mathbf{C} \varepsilon_\varepsilon : \varepsilon_\varepsilon \, d\mathbf{x}$ in $\mathcal{O}$ may be estimated with $\varphi(\mathbf{x}, \mathbf{k}) \equiv \rho(\mathbf{x}) \Gamma(\mathbf{x}, \mathbf{k})$:

$$\lim_{\varepsilon \rightarrow 0} \mathcal{V}_\varepsilon(t) = \frac{1}{2} \int_{\mathcal{O} \times \mathbb{R}^3} \rho(\mathbf{x}) \Gamma(\mathbf{x}, \mathbf{k}) : \mathbf{W}[\mathbf{u}_\varepsilon(\cdot, t)](d\mathbf{x}, d\mathbf{k}), \quad (15)$$

up to some possible boundary effects on $\partial \mathcal{O}$. Similarly, the kinetic energy $\mathcal{T}_\varepsilon(t) := \frac{1}{2} \int_{\mathcal{O}} \rho |\partial_t \mathbf{u}_\varepsilon|^2 \, d\mathbf{x}$ is estimated by:

$$\lim_{\varepsilon \rightarrow 0} \mathcal{T}_\varepsilon(t) = \frac{1}{2} \int_{\mathcal{O} \times \mathbb{R}^3} \rho(\mathbf{x}) \text{Tr} \mathbf{W}[\partial_t \mathbf{u}_\varepsilon(\cdot, t)](d\mathbf{x}, d\mathbf{k}). \quad (16)$$

Now multiplying the elastic wave equation (5) by $(i\varepsilon)^2$ and using the notation of Eq. (12), the latter reads:

$$\mathbf{0} = \rho(\mathbf{x}) ((\varepsilon \mathbf{D}_t)^2 - \Gamma(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x})) \mathbf{u}_\varepsilon + i\varepsilon \mathbf{H}_1(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x}) \mathbf{u}_\varepsilon = (\mathbf{H}_0(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x}, \varepsilon \mathbf{D}_t) + i\varepsilon \mathbf{H}_1(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x})) \mathbf{u}_\varepsilon, \quad (17)$$

where $\mathbf{H}_0(\mathbf{x}, \mathbf{k}, \omega) = \rho(\mathbf{x})(\omega^2 \mathbf{I} - \Gamma(\mathbf{x}, \mathbf{k}))$ is the dispersion matrix and $\mathbf{H}_1(\mathbf{x}, \mathbf{k}) \mathbf{U} = (\nabla_\mathbf{x} \mathbf{C}(\mathbf{x}) : \mathbf{U} \otimes \mathbf{k}) \cdot \nabla_\mathbf{k} \mathbf{k}$ so that $\mathbf{H}_1(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x})$ is a first-order partial differential operator. It should be observed that $\mathbf{H}_1 = \mathbf{0}$ in an homogeneous medium. Our purpose is then to derive the Wigner limit of the solution $\mathbf{u}_\varepsilon$ of the wave equation and how it behaves, because it is needed in (15) and (16) for the computation of the strain and kinetic energies. Considering the scalar product of the wave equation with $\mathbf{u}_\varepsilon$ in $L^2(\mathbb{R}^3)$, one arrives at:

$$\left((\mathbf{H}_0(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x}, \varepsilon \mathbf{D}_t) + i\varepsilon \mathbf{H}_1(\mathbf{x}, \varepsilon \mathbf{D}_\mathbf{x})) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon \right)_{L^2} = 0.$$

One is thus tempted to use the above result (13) to get the high-frequency limit $\varepsilon \rightarrow 0$. However this is not so straightforward because the operator within the brackets on the left hand-side of the above scalar product depends on ε through $\mathbf{H}_1$. The subsequent section details how the limit should be obtained rigorously in this case.

3. Wigner measure of high-frequency elastic waves

The analysis presented here is derived from [15], where first-order hyperbolic systems with constant and slowly varying coefficients are addressed, and [16], where arbitrary order hyperbolic systems with slowly varying coefficients are addressed. The dispersion properties of the Wigner measure (13) of elastic waves are derived in Sect. 3.2, and its evolution properties are derived in Sect. 3.3. Here it is shown that its components in the eigenspaces of the acoustic tensor satisfy transport equations, stating that the energy density in that eigenspaces are transported in phase space with celerities corresponding to the associated eigenvalues. The derivation in Sect. 3.2 and Sect. 3.3 is carried on using a spatio-temporal Wigner transform [12, 17] because the wave equation (17) is second order and the techniques developed in [11, 15] for anti-selfadjoint first order systems cannot be applied in this case. For that purpose, one introduces the space-time variable $\mathbf{s} = (\mathbf{x}, t)$, and the dual impulse variable (wave vector $\times$ circular frequency) $\xi = (\mathbf{k}, \omega) \in \mathbb{R}^4$. Some additional results on the composition and adjoint of operators of the form (12) will also be needed; they are summarized in the subsequent section.

3.1. Some rules of semiclassical calculus

The composition of semiclassical operators (12) is given by the following proposition [20, th. 2.7.4]. Let $\mathbf{a}$ and $\mathbf{b}$ be smooth, compactly supported observables, then one can always find a smooth, compactly supported observable φ such that $\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) = \mathbf{a}(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \circ \mathbf{b}(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s})$ where in addition:

$$\varphi(\mathbf{s}, \xi) = \mathbf{a}(\mathbf{s}, \xi) \mathbf{b}(\mathbf{s}, \xi) - i\varepsilon \nabla_\xi \mathbf{a}(\mathbf{s}, \xi) \cdot \nabla_\mathbf{s} \mathbf{b}(\mathbf{s}, \xi) + O(\varepsilon^2). \quad (18)$$

The higher order terms in ε are given in [20, chap. 6]. The following result characterizes the adjoint operator [20, rem. 2.5.7]. Let $\mathbf{a}$ be a smooth, compactly supported observable, then one can find a smooth observable $\mathbf{b}$ such that $\mathbf{b}(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) = (\mathbf{a}(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}))^*$ and furthermore:

$$\mathbf{b}(\mathbf{s}, \xi) = \mathbf{a}^*(\mathbf{s}, \xi) - i\varepsilon \nabla_\xi \cdot \nabla_\mathbf{s} \mathbf{a}^*(\mathbf{s}, \xi) + O(\varepsilon^2). \quad (19)$$

Once again, the higher order terms in ε are given in [20, chap. 6].

3.2. The dispersion properties

We apply the results of the previous sections extending them to the time and space variables. The spatio-temporal Wigner transform [12, 17]:

$$\mathbf{W}_\varepsilon[\mathbf{u}_\varepsilon](\mathbf{s}, \xi) = \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} e^{i\xi \cdot \mathbf{t}} \mathbf{u}_\varepsilon(\mathbf{s} - \varepsilon \mathbf{t}) \otimes \overline{\mathbf{u}_\varepsilon(\mathbf{s})} d\mathbf{t}$$

allows to capture oscillations of frequency $\omega = O(\varepsilon^{-1})$ and wavenumber $|\mathbf{k}| = O(\varepsilon^{-1})$ for the wave field $\mathbf{u}_\varepsilon$. Let φ be a smooth, compactly supported observable defined on $T^*(\mathcal{O} \times \mathbb{R})$. We apply $\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s})$ to Eq. (17) and compute its $L^2(\mathcal{O} \times \mathbb{R})$ scalar product with $\mathbf{u}_\varepsilon$ to get:

$$0 = (\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \circ \mathbf{H}_0(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + i\varepsilon (\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \circ \mathbf{H}_1(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} \quad (20a)$$

$$= ((\varphi \mathbf{H}_0)(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + O(\varepsilon), \quad (20b)$$

invoking the composition rule (18). Here it is implicitly understood that $\mathbf{H}_0$ and $\mathbf{H}_1$ do not depend on t and $\mathbf{H}_1$ does not depend on ω either. Then by the definition of Eq. (13) extended to the spatio-temporal Wigner measure one obtains:

$$\text{Tr} \int_{T^*(\mathcal{O} \times \mathbb{R})} \varphi(\mathbf{s}, \xi) \mathbf{H}_0(\mathbf{s}, \xi) d\mathbf{W}[\mathbf{u}_\varepsilon](\mathbf{s}, \xi) = 0, \quad \forall \varphi. \quad (21)$$

This equality holds for all observables φ , thus $\mathbf{H}_0 \mathbf{W}[\mathbf{u}_\varepsilon] = \mathbf{0}$. Owing to the properties (7) of the acoustic tensor on $\mathcal{X} := \mathcal{O} \times \mathbb{R}_t \times \mathbb{R}_\mathbf{k}^* \times \mathbb{R}_\omega$, one thus has:

$$\sum_{\alpha=1}^R \mathcal{H}_\alpha \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] = \mathbf{0}$$

on that set, where $\mathcal{H}_\alpha(\mathbf{s}, \xi) = \rho(\mathbf{x})(\omega^2 - c_\alpha^2(\mathbf{x}, \mathbf{k}))$ and $\Pi_\alpha = \mathbf{p}_\alpha \otimes \mathbf{p}_\alpha$. Here $R \leq 3$ is the number of different eigenvalues of the acoustic tensor Γ , and $\mathbf{p}_\alpha = (\mathbf{p}_{\alpha,r})_{r=1, \dots, r_\alpha}$ is the family of eigenvectors associated to the eigenvalue c_α of which order of multiplicity is r_α . For isotropic elasticity for example, $R = 2$ with $\alpha = P$ or $\alpha = S$ and $r_P = 1$, $r_S = 2$. It is assumed in the remaining that these multiplicities remain constant in phase space. The R families of eigenvectors $\mathbf{p}_\alpha$, $1 \leq \alpha \leq R$, form an orthonormal basis of $\mathbb{R}^3$: $\mathbf{p}_\alpha^* \mathbf{p}_\beta = \mathbf{p}_\alpha \cdot \mathbf{p}_\beta = \delta_{\alpha\beta} \mathbf{I}_\alpha$, where $\mathbf{I}_\alpha$ is the $r_\alpha \times r_\alpha$ identity matrix. Then Π_α are projectors, implying that each term in the sum above cancels. Thus:

$$\mathcal{H}_\alpha \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta = 0 \text{ on } \mathcal{X}, \quad \forall \alpha, \beta.$$

Likewise, $\mathbf{W}[\mathbf{u}_\varepsilon]$ being hermitian, $\mathcal{H}_\beta \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta = \mathbf{0}$ on $\mathcal{X}$ and consequently $\mathcal{H}_\beta \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta = 0$ on $\mathcal{X}$, $\forall \alpha, \beta$. Taking the difference of these equalities yields:

$$(\mathcal{H}_\alpha - \mathcal{H}_\beta) \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta = 0 \text{ on } \mathcal{X}, \quad \forall \alpha, \beta.$$

But $\mathcal{H}_\alpha \neq \mathcal{H}_\beta$ on $\mathcal{X}$ as soon as $\alpha \neq \beta$, and $\Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta = \mathbf{0}$ in this case. Expanding $\mathbf{W}[\mathbf{u}_\varepsilon]$ on $\mathcal{X}$ on the basis $(\Pi_\alpha)_\alpha$ such that $\sum_{\alpha=1}^R \Pi_\alpha = \mathbf{I}$:

$$\mathbf{W}[\mathbf{u}_\varepsilon] = \sum_{\alpha=1}^R \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] = \sum_{\alpha, \beta=1}^R \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\beta,$$

the expansion on $\mathcal{X}$ finally reduces to the diagonal terms $\alpha = \beta$ solely:

$$\mathbf{W}[\mathbf{u}_\varepsilon] = \sum_{\alpha=1}^R \mathbf{W}_\alpha[\mathbf{u}_\varepsilon], \quad \mathbf{W}_\alpha[\mathbf{u}_\varepsilon] = \Pi_\alpha \mathbf{W}[\mathbf{u}_\varepsilon] \Pi_\alpha, \quad (22)$$

where $\text{supp } \mathbf{W}_\alpha[\mathbf{u}_\varepsilon] \subset \{(\mathbf{s}, \xi) \in \mathcal{X}; \mathcal{H}_\alpha(\mathbf{s}, \xi) = 0\}$. So the Wigner measure $\mathbf{W}[\mathbf{u}_\varepsilon]$ of the ε -oscillating elastic waves fields $(\mathbf{u}_\varepsilon)$ is expanded into the finite sum of its orthogonal projections onto the different energy paths of the propagation operator $\mathbf{H}_0$. These paths are determined by the equation $\mathcal{H}_\alpha(\mathbf{s}, \xi) = 0$, $1 \leq \alpha \leq R$, in phase space. They correspond to the rays for the medium arising in classical Hamiltonian dynamics.

3.3. The evolution properties

The evolution properties of the projected Wigner measures $\mathbf{W}_\alpha[\mathbf{u}_\varepsilon]$ are derived from the first order terms in (20). Let φ be again a smooth, compactly supported observable defined on $T^*(\mathcal{O} \times \mathbb{R})$, then apply $\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s})$ to Eq. (17), subtract the adjoint equation, and compute its $L^2(\mathcal{O} \times \mathbb{R})$ scalar product with $\mathbf{u}_\varepsilon$ to get:

$$0 = \frac{1}{\varepsilon} [(\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \circ \mathbf{H}_0(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + i\varepsilon (\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \circ \mathbf{H}_1(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} \quad (23a)$$

$$- (\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{H}_0(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon)_{L^2} - i\varepsilon (\varphi(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{H}_1(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon)_{L^2}] \quad (23b)$$

$$= \frac{1}{\varepsilon} ([\varphi, \mathbf{H}_0](\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} - i(\{\varphi, \mathbf{H}_0\}(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + i([\varphi, \mathbf{H}_1](\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s}) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + O(\varepsilon), \quad (23c)$$

because the wave operator $(\mathbf{H}_0 + i\varepsilon \mathbf{H}_1)(\mathbf{s}, \varepsilon \mathbf{D}_\mathbf{s})$ is formally self-adjoint. In the above $[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$ is the usual Lie bracket, $\{\mathbf{A}, \mathbf{B}\} = \nabla_\xi \mathbf{A} \cdot \nabla_\mathbf{s} \mathbf{B} - \nabla_\xi \mathbf{B} \cdot \nabla_\mathbf{s} \mathbf{A}$ is the Poisson bracket, and we have again used (18). Choosing an observable φ which commutes with $\mathbf{H}_0$, the application of Eq. (13) to the space-time Wigner measure yields:

$$\text{Tr} \int_{T^*(\mathcal{O} \times \mathbb{R})} (\{\varphi, \mathbf{H}_0\}(\mathbf{s}, \xi) - [\varphi, \mathbf{H}_1]) d\mathbf{W}[\mathbf{u}_\varepsilon](\mathbf{s}, \xi) = 0, \quad \forall \varphi \text{ such that } [\varphi, \mathbf{H}_0] = \mathbf{0}. \quad (24)$$

Taking a scalar observable $\varphi = \varphi \mathbf{I}$ for example where $\text{supp } \varphi \subset \mathcal{X}$, one has $\text{Tr} \int_{T^*(\mathcal{O} \times \mathbb{R})} \{\varphi, \mathbf{H}_0\} d\mathbf{W}[\mathbf{u}_\varepsilon] = 0$. But:

$$\text{Tr}(\{\varphi, \mathbf{H}_0\} \mathbf{W}[\mathbf{u}_\varepsilon]) = \sum_{\alpha=1}^R \text{Tr}(\mathbf{p}_\alpha^* \{\varphi, \mathbf{H}_0\} \mathbf{p}_\alpha \mathbf{w}_\alpha),$$

where $\mathbf{w}_\alpha := \mathbf{p}_\alpha^* \mathbf{W}[\mathbf{u}_\varepsilon] \mathbf{p}_\alpha$ on $\mathcal{X}$. These matrix measures characterize the energy density and polarizations of the eigenmode α among the R possible ones. Also:

$$(\partial_{s_j} \mathbf{H}_0) \mathbf{p}_\alpha = \partial_{s_j} (\mathbf{H}_0 \mathbf{p}_\alpha) - \mathbf{H}_0 \partial_{s_j} \mathbf{p}_\alpha = \partial_{s_j} (\mathcal{H}_\alpha \mathbf{p}_\alpha) - \mathbf{H}_0 \partial_{s_j} \mathbf{p}_\alpha = (\partial_{s_j} \mathcal{H}_\alpha) \mathbf{p}_\alpha + (\mathcal{H}_\alpha \mathbf{I} - \mathbf{H}_0) \partial_{s_j} \mathbf{p}_\alpha, \quad (25)$$

thus $\mathbf{p}_\beta^* (\partial_{s_j} \mathbf{H}_0) \mathbf{p}_\alpha = \delta_{\alpha\beta} (\partial_{s_j} \mathcal{H}_\alpha) \mathbf{I}_\alpha + (\mathcal{H}_\alpha - \mathcal{H}_\beta) \mathbf{p}_\beta^* (\partial_{s_j} \mathbf{p}_\alpha)$, and particularly $\mathbf{p}_\alpha^* (\partial_{s_j} \mathbf{H}_0) \mathbf{p}_\alpha = (\partial_{s_j} \mathcal{H}_\alpha) \mathbf{I}_\alpha$. Likewise:

$$\mathbf{p}_\beta^* (\partial_{\xi_j} \mathbf{H}_0) \mathbf{p}_\alpha = \delta_{\alpha\beta} (\partial_{\xi_j} \mathcal{H}_\alpha) \mathbf{I}_\alpha + (\mathcal{H}_\alpha - \mathcal{H}_\beta) \mathbf{p}_\beta^* (\partial_{\xi_j} \mathbf{p}_\alpha),$$

with for example $\mathbf{p}_\alpha^* (\partial_{\xi_j} \mathbf{H}_0) \mathbf{p}_\alpha = (\partial_{\xi_j} \mathcal{H}_\alpha) \mathbf{I}_\alpha$. Eq. (24) then reads:

$$\sum_{\alpha=1}^R \int_{T^*(\mathcal{O} \times \mathbb{R})} \{\varphi, \mathcal{H}_\alpha\} d(\text{Tr } \mathbf{w}_\alpha) = 0, \quad \forall \varphi.$$

Since $\text{supp } \mathbf{w}_\alpha \cap \text{supp } \mathbf{w}_\beta \subset \{(s, \xi); \mathbf{k} = \mathbf{0}\}$ if $\alpha \neq \beta$, it can be inferred that the positive measure $W_\alpha = \text{Tr } \mathbf{w}_\alpha = \text{Tr } \mathbf{W}_\alpha[\mathbf{u}_\varepsilon]$ satisfies a Liouville equation in the sense of distributions on $\mathcal{X}$:

$$\{\mathcal{H}_\alpha, W_\alpha\} = 0. \quad (26)$$

Choosing now $\varphi = \mathbf{p}_\beta^* \varphi_\beta \mathbf{p}_\beta$ where $\text{supp } \varphi_\beta \subset \mathcal{X}$ it can be shown that:

$$\text{Tr}(\{(\varphi, \mathbf{H}_0) - (\varphi, \mathbf{H}_1)\} \mathbf{W}[\mathbf{u}_\varepsilon]) = \text{Tr}(\{\varphi_\beta, \mathcal{H}_\beta\} \mathbf{w}_\beta) - \text{Tr}(\mathbf{w}_\beta \varphi_\beta \mathbf{N}_\beta - \mathbf{N}_\beta \varphi_\beta \mathbf{w}_\beta)$$

where $\mathbf{N}_\beta = \mathbf{M}_\beta + \mathbf{p}_\beta^* \mathbf{H}_1 \mathbf{p}_\beta$ with $\mathbf{M}_\beta = \mathbf{p}_\beta^* (\nabla_\xi \mathbf{H}_0 \cdot \nabla_s \mathbf{p}_\beta - \nabla_\xi \mathbf{p}_\beta \cdot \nabla_s \mathcal{H}_\beta)$. Eq. (24) thus reads:

$$\text{Tr} \int_{T^*(\mathcal{O} \times \mathbb{R})} (\{\varphi_\beta, \mathcal{H}_\beta\} d\mathbf{w}_\beta - [d\mathbf{w}_\beta, \varphi_\beta \mathbf{N}_\beta]) = 0, \quad \forall \varphi_\beta, \forall \beta = 1, \dots, R,$$

which shows that the matrix measure $\mathbf{w}_\alpha$ satisfies a matrix transport equation:

$$\{\mathcal{H}_\alpha, \mathbf{w}_\alpha\} = [\mathbf{N}_\alpha, \mathbf{w}_\alpha] \quad (27)$$

in the sense of distributions on $\mathcal{X}$. The Lie bracket involving the rotation matrix $\mathbf{N}_\alpha$ vanishes if the background medium is homogeneous, such that $\mathbf{H}_1 \equiv \mathbf{0}$. It characterizes the rotation of the matrix measures $\mathbf{w}_\alpha$ on the energy rays due to the smooth heterogeneity of the background medium.

4. Radiative transport of the Wigner measure in random media

In this section we briefly outline the main results on the propagation of high-frequency waves in a random anisotropic medium. The so-called weak coupling regime is considered, whereby propagation distances are large compared to the wavelength ε and the perturbations of the elasticity tensor of the background medium are weak and vary at the same scales as the wavelength (meaning that their correlation lengths scale as ε). The subsequent analysis is derived from [12] where scalar (acoustic) waves are considered, and [21] where general first-order anti-selfadjoint systems are considered.

In this setting it is assumed that the elasticity tensor reads [13]:

$$\mathbf{C}(\mathbf{x}) = \mathbf{C}_0^{\frac{1}{2}} \left[\mathbb{I} + \sqrt{\varepsilon} \mathbf{G}_1 \left(\frac{\mathbf{x}}{\varepsilon} \right) \right] \mathbf{C}_0^{\frac{1}{2}}, \quad (28)$$

where $\mathbb{I}$ is the identity tensor, $\mathbf{C}_0$ is the constant elasticity tensor of the isotropic background medium, and $\mathbf{G}_1$ characterizes its fluctuation at the scale ε . This fluctuation is modeled by a tensor-valued, second-order stochastic field $(\mathbf{G}_1(\mathbf{y}); \mathbf{y} \in \mathbb{R}^3)$ which has mean zero and is homogeneous. The latter property means that the cross-correlations of the perturbations at two different locations $\mathbf{y}_1$ and $\mathbf{y}_2$ depend on $\mathbf{y}_1 - \mathbf{y}_2$ solely; it is referred to as anisotropy in the geophysical literature [22]. If the cross-correlations depend on $|\mathbf{y}_1 - \mathbf{y}_2|$ the medium is statistically isotropic, but this does not preclude it from being anisotropic. At last, the inhomogeneities are small as expressed by their $O(\sqrt{\varepsilon})$ amplitude. This size is the unique scaling which allows them to significantly modify the energy spreading in the transport regime

at long propagation distances [12]. Larger fluctuations would lead to localization of the waves, a situation beyond the scope of kinetic models.

Considering this rescaling of the elasticity tensor, $\mathbf{H}_1 \equiv \mathbf{0}$ and Eq. (20) is modified to:

$$O(\varepsilon^{\frac{3}{2}}) = (\varphi \circ \mathbf{H}_0(\mathbf{x}, t, \varepsilon \mathbf{D}_x + \mathbf{D}_y, \varepsilon \mathbf{D}_t) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + \sqrt{\varepsilon} (\varphi \circ \mathbf{H}_\varepsilon(\mathbf{y}, \varepsilon \mathbf{D}_x + \mathbf{D}_y) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2},$$

where $\mathbf{H}_\varepsilon(\mathbf{y}, \mathbf{k}) \mathbf{U} := (\mathbf{G}_1(\mathbf{y}) : \mathbf{U} \otimes \mathbf{k}) \mathbf{k}$ and $\mathbf{y} \equiv \frac{\mathbf{x}}{\varepsilon}$. In the above $\varepsilon \mathbf{D}_x$ is replaced by $\varepsilon \mathbf{D}_x + \mathbf{D}_y$, as $\mathbf{x}$ and $\mathbf{y}$ are seen as independent variables. Taking the limit $\varepsilon \rightarrow 0$ the same result as in the deterministic case is derived, namely Eq. (21). The Wigner measure of the elastic waves has the same dispersion properties and spectral expansion (22), which is expected from the choice of the perturbation scaling (28). However the situation is different for the evolution properties. Indeed Eq. (23) is modified to:

$$O(\sqrt{\varepsilon}) = \frac{1}{\varepsilon} ([\varphi, \mathbf{H}_0](\mathbf{s}, \varepsilon \mathbf{D}_s) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} - i (\{\varphi, \mathbf{H}_0\}(\mathbf{s}, \varepsilon \mathbf{D}_s) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} + \frac{i}{\sqrt{\varepsilon}} (\mathcal{L}_\varepsilon(\mathbf{y}, \varepsilon \mathbf{D}_x + \mathbf{D}_y) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2},$$

where $\mathcal{L}_\varepsilon$ is a linear operator arising from the difference of $\mathbf{H}_\varepsilon$ and its adjoint. Then it is shown in [11, 12, 21] that:

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\sqrt{\varepsilon}} \mathbb{E} \{ (\mathcal{L}_\varepsilon(\mathbf{y}, \varepsilon \mathbf{D}_x + \mathbf{D}_y) \mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon)_{L^2} \} = \int_{T^*(\mathcal{O} \times \mathbb{R})} (\varphi \mathcal{Q})(\mathbf{s}, \xi) \mathbb{E} \{ \mathbf{W}[\mathbf{u}_\varepsilon] \} (\mathbf{d}\mathbf{s}, \mathbf{d}\xi), \quad \forall \varphi,$$

where $\mathbb{E} \{ \cdot \}$ stands for the mathematical expectation, and $\mathcal{Q}$ is a macroscopic integral operator described in detail in [21]. This limit holds true in return for a mixing assumption, allowing to separate the microscopic and macroscopic scales in a statistical sense: $\mathbb{E} \{ f(\mathbf{y}_1) f^*(\mathbf{y}_2) W(\mathbf{x}) \} \simeq R_f(\mathbf{y}_1 - \mathbf{y}_2) \mathbb{E} \{ W \}(\mathbf{x})$ where R_f is the correlation function of the random field $(f(\mathbf{y}); \mathbf{y} \in \mathbb{R}^3)$ with correlation length ε , *e.g.* the fluctuation tensor $\mathbf{G}_1$ in the present case. Consequently, the matrix transport equations (27) are modified to:

$$\{ \mathcal{H}_\alpha, \mathbb{E} \{ \mathbf{w}_\alpha \} \} = \sum_{\beta=1}^R \mathcal{Q}_{\alpha\beta} \mathbb{E} \{ \mathbf{w}_\beta \}, \quad 1 \leq \alpha \leq R, \quad (29)$$

with $\mathcal{Q}_{\alpha\beta} = \mathbf{p}_\alpha^* \mathcal{Q} \mathbf{p}_\beta$. The latter are so-called collision operators which contribute to mix the polarization states $1 \leq \alpha \leq R$ of the high-frequency waves through their scattering on the random inhomogeneities of the background medium. The left-hand side of Eq. (29) corresponds to pure transport of the average energy density along the rays of the background medium, as in Eq. (27). Eq. (29) is a matrix radiative transfer equation, or linear Boltzmann equation, which is well known to evolve into an irreversible diffusion equation after long times; see for example [5, 11] and references therein. The main interests of these results are twofold. First, a quantitative expression of the scattering effects of inhomogeneities in an anisotropic random medium is established. Second, they justify the derivation of an irreversible, deterministic dynamic system of Boltzmann type from a reversible, stochastic wave system. Actually the latter property extends to general linear dynamical systems as demonstrated in [21].

5. Conclusions

In this paper we have outlined the derivation of kinetic models for mesoscopic wave propagation phenomena at ranges between the small wavelength and some large propagation distances with respect to the transport mean free paths. The proposed model considers energy densities in phase space to account for the spreading of waves in many directions at each point in physical space due to their multiple scattering on inhomogeneities when the latter vary at the same scale as the wavelength. These scattering phenomena occur in addition to pure transport of the energy density along the rays of the background medium in classical Hamiltonian dynamics. They are crucial for the emergence of a diffusion regime of waves when they have lost the memory of their polarization features and propagation directions. In this respect, a detailed analysis of the influence of material anisotropy on the diffusion operator remains to be done, both at the microscopic level and statistical level; it is the subject of ongoing investigations. As shown by the consideration of interface and boundary conditions for the high-frequency vibrations of slender structures, power flow reflection/transmission phenomena are very much likely to accelerate the onset of diffuse waves. The role of boundary conditions for anisotropic media shall also be investigated in future researches.

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